

# Linus Adzanku

## Mechanical Engineer

**Email** | (+44) 7801906656 | **LinkedIn** | **Portfolio**

### Professional Summary

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Mechanical Engineer with an MEng in Aerospace Engineering and project experience across mechanical design, CAD, FEA, CFD and testing & validation. Experienced using SolidWorks, ANSYS and MATLAB to develop and analyse physical engineering systems, from requirements and first-principles analysis through design, simulation, manufacture and test.

### Skills

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**Design & Development:** Mechanical design, CAD, SolidWorks, engineering drawings, prototyping, design for manufacture, requirements definition

**Analysis & Simulation:** FEA, CFD, ANSYS Mechanical, Fluent & CFX, MATLAB, Simulink, engineering calculations

**Testing & Validation:** Test planning, validation, risk assessment, leak testing, data analysis, FFT/signal processing

**Programming & Tools:** Python, C++, VBA, Excel, Git/GitHub, Linux, NASA CEA

### Engineering & Research Projects

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#### 500 N Hybrid Rocket Engine — Injector Design, Manufacture & Testing

*MEng R&D Capstone Project*

- Co-led a five-person team developing a 500 N N<sub>2</sub>O/HDPE hybrid rocket engine and owned injector development from preliminary sizing through CAD, manufacture, integration and test preparation.
- Developed the injector using theoretical and experimental assumptions, NASA CEA, MATLAB/Excel and SolidWorks, incorporating manufacturing constraints and technician feedback.
- Supported manufacture and testing, including leak testing and investigation of chamber-sealing issues identified during the test campaign.

#### Cartography UAS

*Personal Project | Ongoing* | [GitHub Link](#)

- Independently developing a fixed-wing UAS from requirements and conceptual design towards detailed design, manufacture and eventual ground and flight testing.
- Defining the system architecture across airframe, propulsion, power, avionics, sensors, communications and embedded-control systems.
- Maintaining requirements, trade studies and engineering decisions using Python, Git/GitHub and structured engineering analysis.

#### VAWT Aeroacoustics

*MEng Research Project*

- Developed transient CFD models of a vertical-axis wind turbine in ANSYS Fluent using Transition SST/SBES to investigate aerodynamic and aeroacoustic behaviour.

- Analysed transient pressure data from a four-receiver array using FFT-based spectral analysis to investigate blade-passing-frequency behaviour, tonal content and acoustic directivity.
- Evaluated simulation quality and interpreted aerodynamic and acoustic trends from the numerical results.

## **Turboprop Aircraft Conceptual Design**

### *University Module Project*

- Contributed to the conceptual design of a passenger turboprop aircraft within a ten-person multidisciplinary engineering team.
- Owned landing-gear design and economic analysis while supporting aerodynamics and weight-and-balance work using MATLAB and engineering calculations.
- Evaluated design trade-offs involving performance, manufacturability, maintenance and airport infrastructure.

## **Professional Experience**

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### **Junior Buyer — Unite Students**

Bristol, United Kingdom | Jun 2025 – Dec 2025

- Managed procurement activities from requirements definition through supplier assessment and recommendation, coordinating internal stakeholders and up to 50 external suppliers.
- Developed MATLAB, VBA and Excel tools to process supplier and commercial data and support evidence-based decision-making.
- Managed procurement of a digital-signage system from requirements capture through supplier comparison and implementation, delivering the solution two weeks ahead of schedule.

### **Procurement Intern — Unite Students**

Bristol, United Kingdom | Jun 2024 – Aug 2024

- Audited 30+ supplier contracts against service requirements and supported tender evaluation through supplier benchmarking and data analysis.
- Used Excel, PivotTables and VLOOKUP to convert commercial and performance data into reports for procurement and facilities stakeholders.

## **Education**

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**University of the West of England**, Bristol, England

*MEng Aerospace Engineering | Grade: Merit*

*Oct 2021 – Aug 2025*

Relevant study: structural mechanics, materials and manufacturing, aerodynamics, aeroacoustics, aircraft systems, propulsion, aircraft design and flight testing.

## **Additional Information**

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**Portfolio:** <https://nusnaaa.github.io/>

**Languages:** English — Fluent | Japanese — Elementary